

```
figure.SpringBootApplication;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@SpringBootApplication
public class DemoApplication {

    public static void main(String[] args) {
        SpringApplication.run(DemoApplication.class, args);
    }

    @RestController
    class HelloController {

        @GetMapping("/")
        public String hello() {
            return "Hello, World!";
        }
    }
}
```

2) Android Development:

Java was the official language for Android development before Kotlin. Many Android applications are still developed in Java.

Example: A simple Android activity

```
package com.example.myfirstapp;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

3) Enterprise Applications:

Java is commonly used for developing enterprise-level applications, including financial services, e-commerce, and customer relationship management (CRM) systems.

Example: Java Enterprise Edition (Java EE) with JPA and EJB

```
// Entity class
import javax.persistence.Entity;
```

```
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;

@Entity
public class User {
    @Id
    @GeneratedValue(strategy = GenerationType.AUTO)
    private Long id;
    private String name;

    // Getters and setters
}

// EJB for business logic
import javax.ejb.Stateless;
import javax.persistence.EntityManager;
import javax.persistence.PersistenceContext;

@Stateless
public class UserBean {
    @PersistenceContext
    private EntityManager em;

    public void createUser(String name) {
        User user = new User();
        user.setName(name);
        em.persist(user);
    }
}
```

4) Big Data:

Java is used in big data technologies like Apache Hadoop and Apache Spark, providing robust solutions for data processing and analytics.

Example: A simple Hadoop MapReduce job

```
// Mapper class
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

import java.io.IOException;

public class WordCountMapper extends Mapper<Object, Text, Text, IntWritable> {
    private final static IntWritable one = new IntWritable(1);
    private Text word = new Text();

    public void map(Object key, Text value,
```